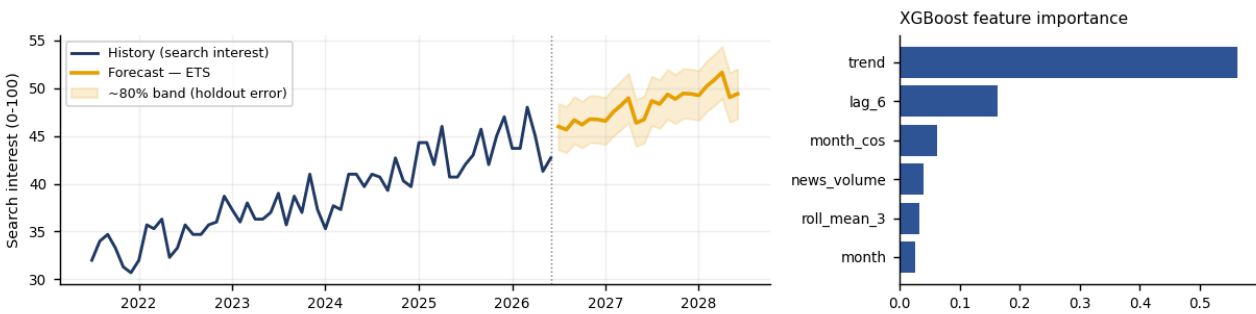


Fari — 24-month demand forecast

Target: monthly search-interest momentum (Week 5). Backtest on the last 12 months; best model by MASE: **ETS** (MAPE 4.1%, MASE 1.00).

Fari (eldercare) — 24-month demand forecast



Model comparison (backtest)

Model	MAPE	MASE	12m	24m
Seasonal naive	6.1%	1.52	43	43
ETS *	4.1%	1.00	47	49
ARIMA	4.9%	1.22	44	44
Prophet	4.6%	1.12	47	49
XGBoost	4.3%	1.06	43	43

Point forecast & drivers

12-month point estimate **47**, 24-month **49** (search-interest index, ~80% band shown from holdout error). Top XGBoost drivers: **trend, lag_6, month_cos**.

Sanity-check vs published analyst forecasts

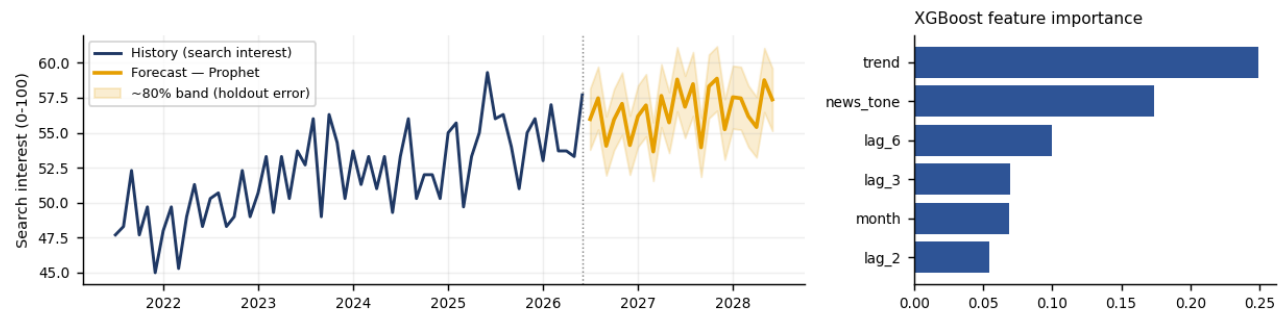
Eldercare/companion-robot adoption is widely expected to grow but is penetration-constrained; our near-flat-to-up momentum reflects that. No single analyst point forecast is anchored.

Method: search-momentum target from Week 5; models statsmodels/Prophet/XGBoost; MASE vs in-sample naive-1. Forecasts are a relative demand read, not absolute unit/market forecasts; validate against internal data before planning. Prepared by Ziheng Wang, Week 10.

Senpai — 24-month demand forecast

Target: monthly search-interest momentum (Week 5). Backtest on the last 12 months; best model by MASE: **Prophet** (MAPE 3.0%, MASE 0.54).

Senpai (education) — 24-month demand forecast



Model comparison (backtest)

Model	MAPE	MASE	12m	24m
Seasonal naive	4.2%	0.75	58	58
ETS	3.1%	0.56	58	60
ARIMA	3.8%	0.68	55	55
Prophet *	3.0%	0.54	59	57
XGBoost	4.0%	0.72	56	56

Point forecast & drivers

12-month point estimate **59**, 24-month **57** (search-interest index, ~80% band shown from holdout error). Top XGBoost drivers: **trend, news_tone, lag_6**.

Sanity-check vs published analyst forecasts

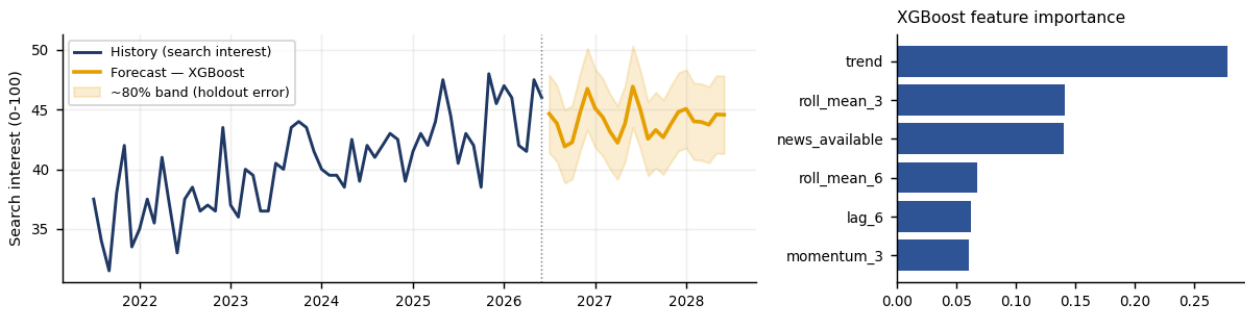
Educational-robotics forecasts project steady growth tied to budget cycles; our stable upward momentum is consistent. Treated as a relative read.

Method: search-momentum target from Week 5; models statsmodels/Prophet/XGBoost; MASE vs in-sample naive-1. Forecasts are a relative demand read, not absolute unit/market forecasts; validate against internal data before planning. Prepared by Ziheng Wang, Week 10.

Sentinel Prime AI — 24-month demand forecast

Target: monthly search-interest momentum (Week 5). Backtest on the last 12 months; best model by MASE: **XGBoost** (MAPE 5.7%, MASE 0.96).

Sentinel Prime AI (indoor_security) — 24-month demand forecast



Model comparison (backtest)

Model	MAPE	MASE	12m	24m
Seasonal naive	6.1%	1.06	46	46
ETS	6.4%	1.05	46	48
ARIMA	6.1%	1.04	45	45
Prophet	6.6%	1.08	45	48
XGBoost *	5.7%	0.96	47	45

Point forecast & drivers

12-month point estimate **47**, 24-month **45** (search-interest index, ~80% band shown from holdout error). Top XGBoost drivers: **trend, roll_mean_3, news_available**.

Sanity-check vs published analyst forecasts

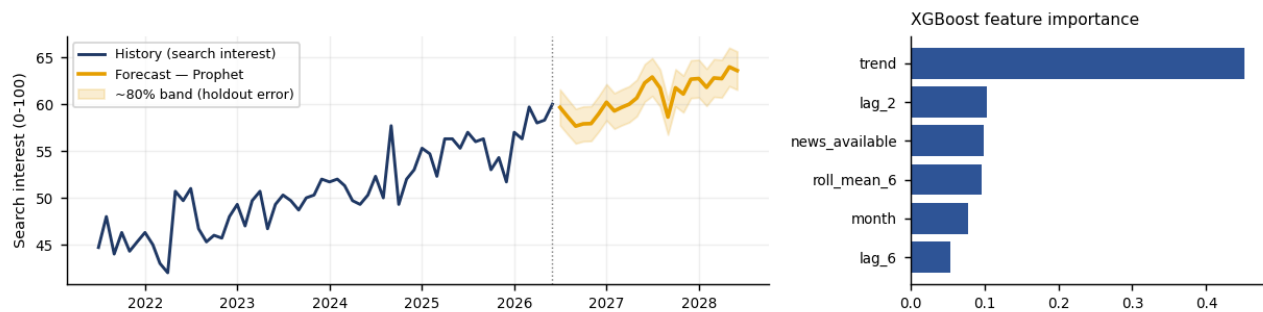
Public service-robotics forecasts broadly project double-digit CAGR for professional security robots; our modest upward search momentum is consistent, though no single audited point forecast is used as an anchor.

Method: search-momentum target from Week 5; models statsmodels/Prophet/XGBoost; MASE vs in-sample naive-1. Forecasts are a relative demand read, not absolute unit/market forecasts; validate against internal data before planning. Prepared by Ziheng Wang, Week 10.

Aido Rover — 24-month demand forecast

Target: monthly search-interest momentum (Week 5). Backtest on the last 12 months; best model by MASE: **Prophet** (MAPE 2.5%, MASE 0.69).

Aido Rover (outdoor_patrol) — 24-month demand forecast



Model comparison (backtest)

Model	MAPE	MASE	12m	24m
Seasonal naive	5.6%	1.56	60	60
ETS	2.7%	0.73	61	64
ARIMA	4.4%	1.23	58	58
Prophet *	2.5%	0.69	62	64
XGBoost	4.4%	1.23	59	59

Point forecast & drivers

12-month point estimate **62**, 24-month **64** (search-interest index, ~80% band shown from holdout error). Top XGBoost drivers: **trend, lag_2, news_available**.

Sanity-check vs published analyst forecasts

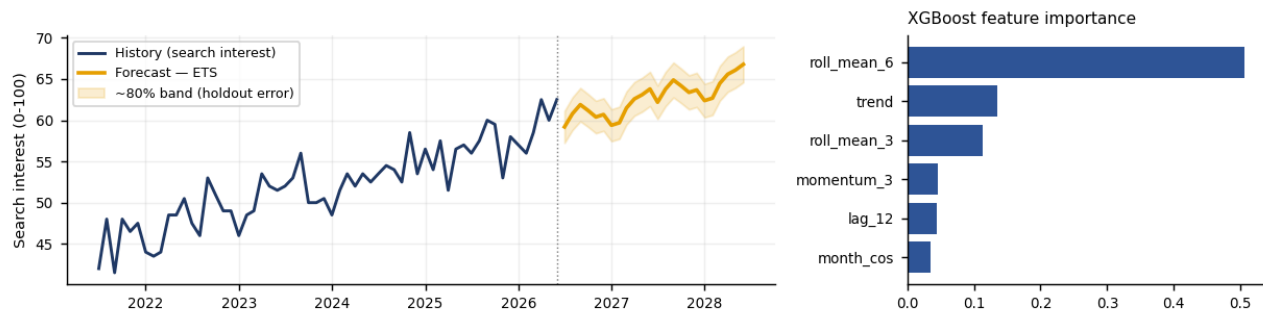
Perimeter/patrol robotics is an emerging professional-service segment; published forecasts are directional. Our forecast is treated as a relative momentum read, not an absolute market size.

Method: search-momentum target from Week 5; models statsmodels/Prophet/XGBoost; MASE vs in-sample naive-1. Forecasts are a relative demand read, not absolute unit/market forecasts; validate against internal data before planning. Prepared by Ziheng Wang, Week 10.

Aido Humanoid — 24-month demand forecast

Target: monthly search-interest momentum (Week 5). Backtest on the last 12 months; best model by MASE: **ETS** (MAPE 2.5%, MASE 0.59).

Aido Humanoid (humanoid) — 24-month demand forecast



Model comparison (backtest)

Model	MAPE	MASE	12m	24m
Seasonal naive	7.3%	1.73	62	62
ETS *	2.5%	0.59	64	67
ARIMA	3.3%	0.77	60	60
Prophet	3.0%	0.71	64	67
XGBoost	6.1%	1.45	61	61

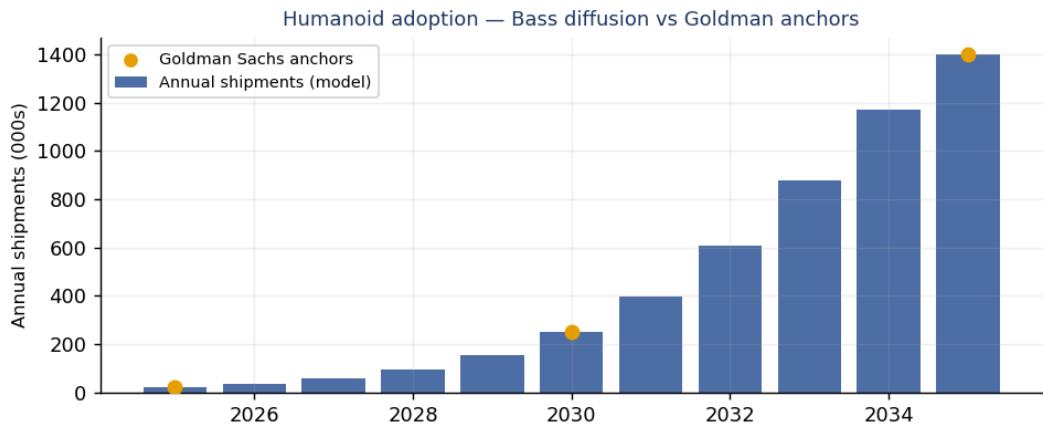
Point forecast & drivers

12-month point estimate **64**, 24-month **67** (search-interest index, ~80% band shown from holdout error). Top XGBoost drivers: **roll_mean_6**, **trend**, **roll_mean_3**.

Sanity-check vs published analyst forecasts

Goldman Sachs (Jan 2024): ~20k humanoid shipments 2025 -> ~250k (2030) -> ~1.4M/yr (2035), ~70% CAGR, \$38B TAM by 2035. Morgan Stanley (Apr 2025): ~13M humanoids in service by 2035. Our search-momentum rise is directionally consistent with this accelerating adoption.

Adoption curve (Bass diffusion)



Bass fit to Goldman anchors ($p=0.0018$ $q=0.520$ $M=11273308$); hits 20k/2025, 250k/2030, 1.4M/2035 with a smooth S-curve; cumulative installed base ~4.3M by 2035 (between Goldman shipment view and Morgan Stanley's 13M-in-service scenario).

Method: search-momentum target from Week 5; models statsmodels/Prophet/XGBoost; MASE vs in-sample naive-1. Forecasts are a relative demand read, not absolute unit/market forecasts; validate against internal data before planning. Prepared by Ziheng Wang, Week 10.